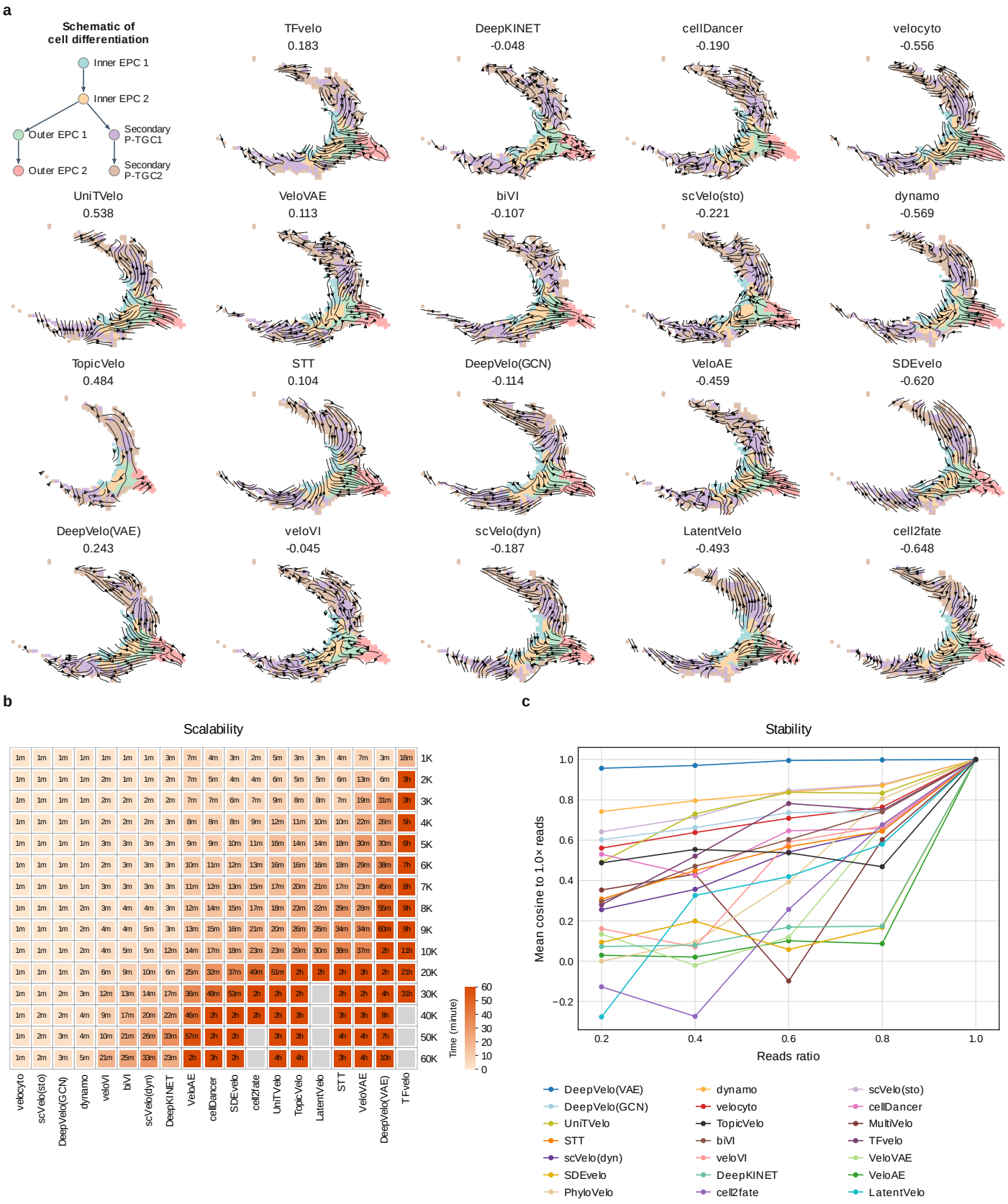

a
Schematic of cell differentiation
Inner EPC 1
Inner EPC 2
Outer EPC 1
Secondary P-TGC1
Outer EPC 2
Secondary P-TGC2
TFvelo 0.183
DeepKINET -0.048
cellDancer -0.190
velocyto -0.556
UniTVelo 0.538
VeloVAE 0.113
biVI -0.107
scVelo(sto) -0.221
dynamo -0.569
TopicVelo 0.484
STT 0.104
DeepVelo(GCN) -0.114
VeloAE -0.459
SDEvelo -0.620
DeepVelo(VAE) 0.243
veloVI -0.045
scVelo(dyn) -0.187
LatentVelo -0.493
cell2fate -0.648
b
Scalability
1K
2K
3K
4K
5K
6K
7K
8K
9K
10K
20K
30K
40K
50K
60K
Time (minute)
velocyto
scVelo(sto)
DeepVelo(GCN)
dynamo
veloVI
biVI
scVelo(dyn)
DeepKINET
VeloAE
cellDancer
SDEvelo
cell2fate
UniTVelo
TopicVelo
LatentVelo
STT
VeloVAE
DeepVelo(VAE)
TFvelo
c
Stability
Mean cosine to 1.0× reads
Reads ratio
DeepVelo(VAE)
DeepVelo(GCN)
UniTVelo
STT
scVelo(dyn)
SDEvelo
PhyloVelo
dynamo
velocyto
TopicVelo
biVI
veloVI
DeepKINET
cell2fate
scVelo(sto)
cellDancer
MultiVelo
TFvelo
VeloVAE
VeloAE
LatentVelo